

Data Stores & Pipeline

Units

3 Credits

Presented by

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Course Description

This course is your gateway to becoming a modern data engineer. You'll learn how data flows through today's enterprises—starting from storage systems and warehouses to real-time pipelines powered by Kafka, Spark, and Airflow. Each concept is brought to life through hands-on labs and real-world case studies from industries like e-commerce, fintech, logistics, and ride-sharing.

By the end of the course, you'll not only understand the theory behind data storage, transformation, orchestration, and governance, but also gain the practical skills to design, automate, and secure scalable data pipelines that drive business decisions in the real world.



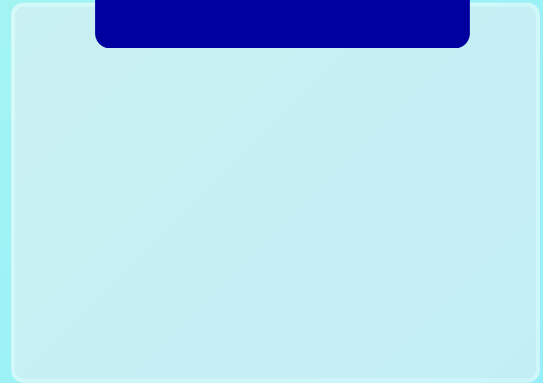
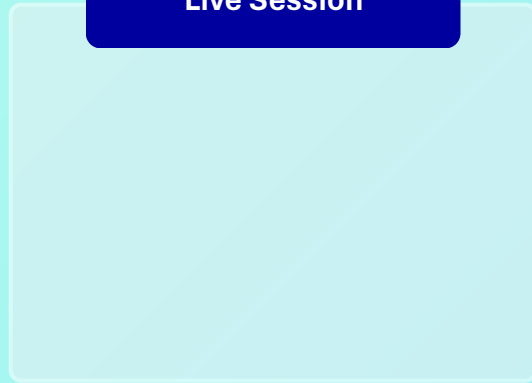
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Week 1 - Foundations of Data Storage Paradigms

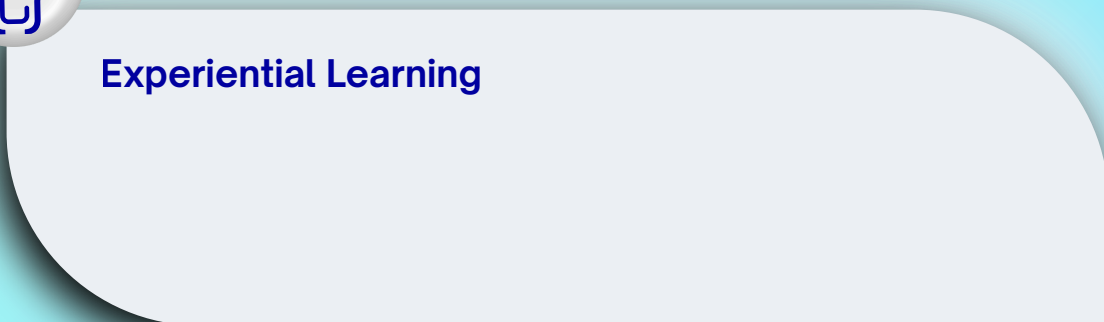
- ◆ Module introduction
- ◆ File systems to RDBMS
- ◆ Data warehouses and BI
- ◆ Data lake to lakehouse
- ◆ OLTP systems
- ◆ OLAP workloads
- ◆ OLTP vs OLAP use cases
- ◆ Overview of RDBMS
- ◆ Columnar databases explained
- ◆ When to use what?
- ◆ Deep dive into ACID
- ◆ BASE and distributed systems
- ◆ Ride-sharing app use case overview



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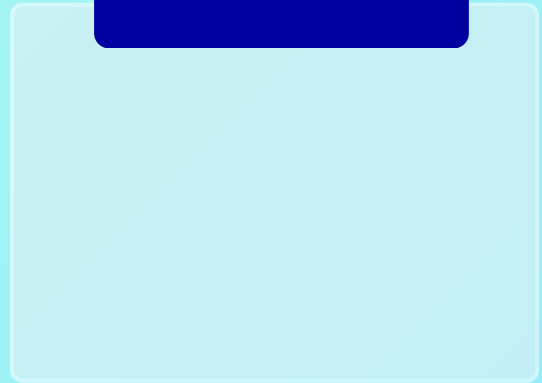
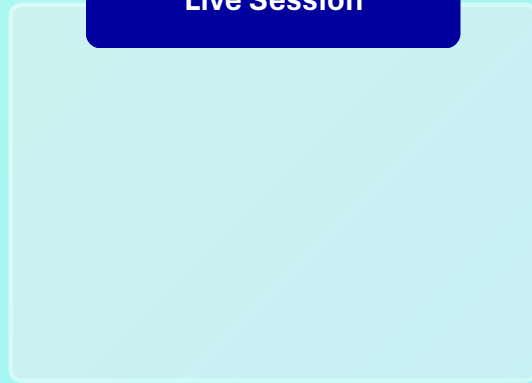
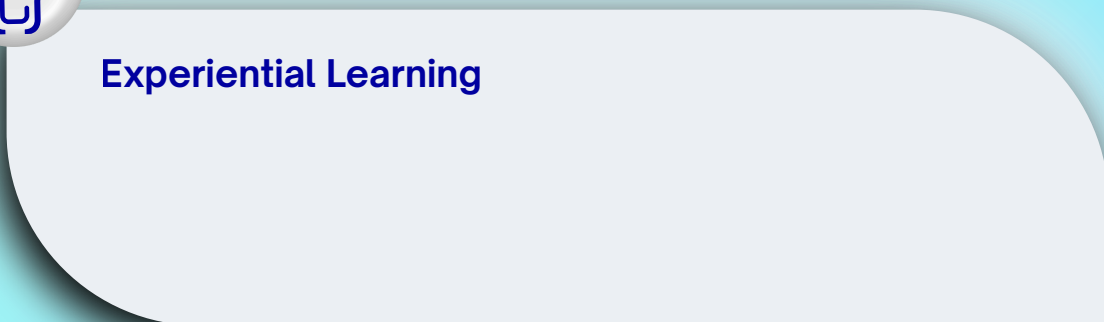
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2

Week 2 - Modern Data Warehousing

- ◆ Module introduction
- ◆ What makes a warehouse “modern”?
- ◆ Key functional features
- ◆ Integrating with data lakes
- ◆ What is MPP and why it matters
- ◆ How queries are parallelized
- ◆ Overview of Snowflake
- ◆ Snowflake demo
- ◆ Overview of Amazon Redshift
- ◆ Redshift demo
- ◆ Financial services reporting & analytics use case overview

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Week 3 - Data Lakes and Lakehouse Foundations

- ◆ Module introduction
- ◆ What is a data lake?
- ◆ What is a lakehouse?
- ◆ Medallion architecture – bronze, silver, gold layers
- ◆ Traditional CSV and its limitations
- ◆ Introduction to Parquet and Avro
- ◆ Format comparison and benchmarks
- ◆ What is cloud object storage?
- ◆ What is Spark?
- ◆ Spark features and advantages
- ◆ Clickstream analytics for e-commerce use case overview



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Week 4 - Data Infrastructure in the Enterprise

- ◆ Module introduction
- ◆ Data stack and its key components
- ◆ Monolith to microservices
- ◆ Distributed data processing
- ◆ On-prem infrastructure
- ◆ Cloud-native environments
- ◆ Hybrid and multi-cloud
- ◆ AWS data ecosystem
- ◆ GCP data ecosystem
- ◆ Azure data ecosystem
- ◆ Logistics company on AWS use case overview

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5

Week 5 - Modern Table Formats – Iceberg, Hudi, Delta Lake

- ◆ Module introduction
- ◆ Why modern table formats?
- ◆ Overview: Iceberg
- ◆ Overview: Delta Lake
- ◆ Overview: Apache Hudi
- ◆ Table format comparison matrix
- ◆ Data storage in Spark
- ◆ What is RDD & DataFrame?
- ◆ Transformations & actions
- ◆ Demo: transformations and actions
- ◆ Uber's Apache Hudi use case overview

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6

Week 6 - Standard Data Pipeline Architecture

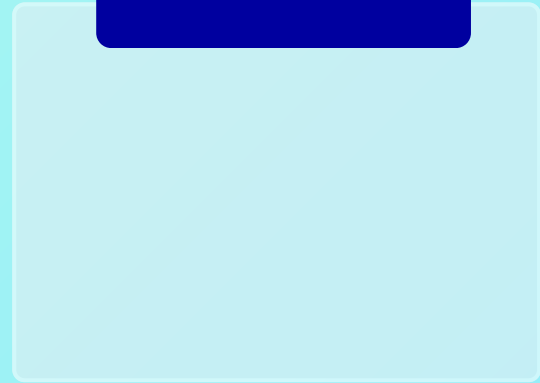
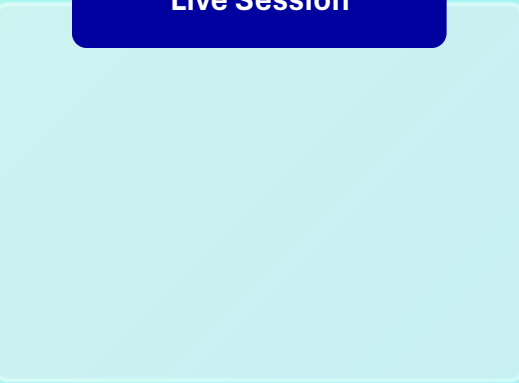
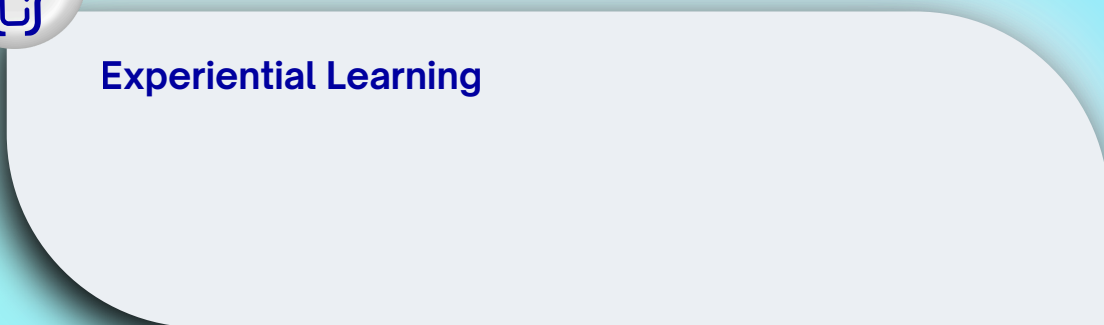
- ◆ Module introduction
- ◆ Batch ingestion
- ◆ Micro-batch and real-time ingestion
- ◆ Ingestion strategy comparison
- ◆ What is CDC and why it matters
- ◆ CDC tools and approaches
- ◆ Lambda architecture
- ◆ Kappa architecture
- ◆ When to use what?
- ◆ Introduction to event-driven architecture
- ◆ Tools: Kafka, GCP Pub/Sub, AWS Kinesis
- ◆ Online food delivery use case overview

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7

Week 7 - Batch Data Ingestion

- ◆ Module introduction
- ◆ Common file formats
- ◆ Reading files using PySpark
- ◆ Optimizing file reads
- ◆ Demo: reading files using PySpark
- ◆ JDBC-based ingestion using PySpark
- ◆ Demo: reading RDBMS tables using PySpark
- ◆ Calling REST APIs
- ◆ Persisting API data
- ◆ Demo: calling REST APIs using PySpark
- ◆ Common validation rules
- ◆ Error handling techniques
- ◆ Resilient batch ingestion with PySpark
- ◆ PySpark use case overview

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Week 8 - ELT vs. ETL Methodologies

- ◆ Module introduction
- ◆ What is ETL?
- ◆ What is ELT?
- ◆ ETL architecture overview
- ◆ ELT architecture with cloud data warehouses
- ◆ ETL vs. ELT – which one and when?
- ◆ ETL pipeline with PySpark – use case overview
- ◆ Demo: building ETL with PySpark
- ◆ ELT with dbt and Snowflake – use case overview
- ◆ ELT with dbt and Snowflake – architecture overview

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Week 9 - Data Transformation, Processing, and Data Quality

- ◆ Module introduction
- ◆ Handling missing and null values
- ◆ Outlier detection and removal
- ◆ Data type casting and normalization
- ◆ Different joins in Spark
- ◆ Skew handling in joins
- ◆ Partitioning for scalable joins
- ◆ Defining quality metrics
- ◆ Writing validation rules
- ◆ Logging and alerting bad records
- ◆ Overview of DQ frameworks
- ◆ Retail banking pipeline use case overview



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Week 10 - Streaming Data Ingestion

- ◆ Module introduction
- ◆ Introduction to Kafka
- ◆ Kafka architecture
- ◆ Kafka fundamentals
- ◆ Demo on writing a Kafka producer
- ◆ Demo on writing a Kafka consumer to read messages
- ◆ Introduction to Spark streaming concepts
- ◆ Structured streaming API with Kafka integration
- ◆ Transforming and outputting data to console/parquet
- ◆ Twitter data pipeline use case overview



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Week 11 - Orchestrating Data Pipelines with Apache Airflow

- ◆ Module introduction
- ◆ What is orchestration?
- ◆ Tools overview
- ◆ What is Airflow?
- ◆ Core concepts: DAGs, operators, tasks
- ◆ Demo on writing a basic DAG
- ◆ Advanced task features
- ◆ Monitoring and debugging in
- ◆ Airflow UI
- ◆ Marketing campaign orchestration use case overview

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12

Week 12 - CI/CD Integration & Automation in Data Pipelines

- ◆ Module introduction
- ◆ Git basics
- ◆ GitHub for data projects
- ◆ What is CI/CD in data engineering
- ◆ Pipeline stages (build, test, deploy)
- ◆ GitHub Actions
- ◆ Jenkins and GitLab CI
- ◆ When to use what
- ◆ Unit testing with pytest
dbt tests for data quality
- ◆ CI/CD for data pipelines use case overview



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Week 13 - Data Governance, Security, and Monitoring in Data Pipelines

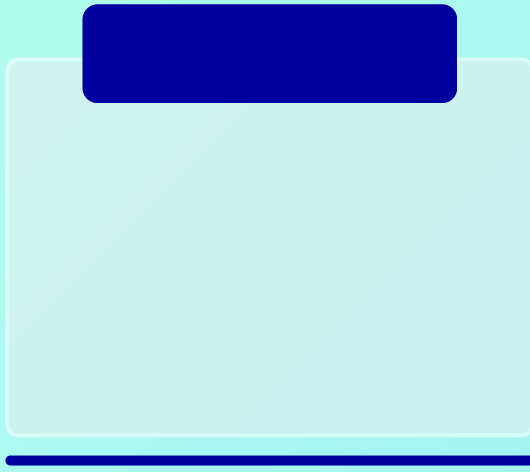
- ◆ Module introduction
- ◆ What is data governance & why it matters
- ◆ GDPR & CCPA overview
- ◆ Engineering responsibilities under compliance
- ◆ Data masking techniques
- ◆ Encryption in data pipelines
- ◆ Access control mechanisms
- ◆ Metadata & governance
- ◆ Lineage: the audit trail of data
- ◆ Tools for metadata & lineage
- ◆ Why monitoring pipelines is critical
- ◆ Prometheus for metric collection
- ◆ Grafana dashboards & alerts
- ◆ Securing PII in fintech pipelines use case overview

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14-15

Comprehensive examination



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Congratulations